

Positive completion preserving a fixed number of prefix moments

Complete proof; three independent mathematical reviews completed

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Fix an integer $q \geq 0$. Constants with subscript q may depend on q , which does not grow with the number $m \geq \max(1, q)$ of polynomial degrees. Let $0 = b_0 < b_1 < \cdots < b_K < b_{K+1} = 1$ and $\Delta = \min_i(b_{i+1} - b_i)$. Put

$$S_s(a) = K^{-1} \sum_{b_i > a} b_i^s, \quad V_m = \text{span}\{a^r : 0 \leq r \leq m\} + \text{span}\{a^r S_s(a) : r + s \leq m\}.$$

Let P_q be orthogonal projection onto polynomials of degree at most q separately on each interval (b_i, b_{i+1}) , and put

$$\mathcal{S}_q = \text{span}\{a^r : 0 \leq r \leq q\} + \text{span}\{a^r S_s(a) : 0 \leq r \leq q, r + s \leq m\}.$$

All norms and projections below use Lebesgue measure.

Lemma 1 (A quotient estimate from matching derivative jumps). *For every $F \in V_m$,*

$$\text{dist}_{L^2}(F, \mathcal{S}_q) \leq C_q(m+1)^{2q+2} \Delta^{-q-1} \|(I - P_q)F\|_2.$$

In particular $V_m \cap \text{ran } P_q = \mathcal{S}_q$.

Proof. The jump of $F^{(\ell)}$ at b_i is $-D_\ell(b_i)/K$, for a polynomial D_ℓ of degree at most $m - \ell$. For $0 \leq \ell \leq q$ and $0 \leq t \leq m - \ell$, define

$$\Phi_{\ell,t}(a) = \frac{1}{\ell!} \sum_{j=0}^{\ell} \binom{\ell}{j} (-1)^{\ell-j} a^j S_{t+\ell-j}(a).$$

Its jump polynomial at b_i is $-b_i^t(a - b_i)^\ell / (K\ell!)$. Thus its derivatives have zero jump in every order except ℓ , where the jump is exactly $-b_i^t/K$; derivatives above ℓ vanish inside every cell. Also $\Phi_{\ell,t} \in \mathcal{S}_q$. Expanding each D_ℓ in powers and adding the corresponding $\Phi_{\ell,t}$ constructs $G \in \mathcal{S}_q$ such that $F - G$ and its first q derivatives are continuous on $[0, 1]$.

Subtract from $F - G$ its Taylor polynomial of degree q at zero. Repeated integration and Cauchy–Schwarz give $\|F - G\|_2 \leq C_q \|F^{(q+1)}\|_2$, after absorbing that polynomial into G . On each cell I , the fixed-order inverse Markov inequality applied to $F - P_q F$ gives

$$\|F^{(q+1)}\|_{L^2(I)} \leq C_q(m+1)^{2q+2} |I|^{-q-1} \|F - P_q F\|_{L^2(I)}.$$

Summing squares proves the estimate. Its zero case gives the kernel identity; the reverse inclusion is immediate. $\square$

Theorem 1 (Higher prefix completion). *Suppose numbers $r_{s,i}$, $0 \leq s \leq q$, $0 \leq i \leq K+1$, satisfy*

$$r_{s,0} = 0, \quad r_{s,K+1} = 1/(s+1),$$

and the compatibility identities

$$K^{-1} \sum_{i=1}^K b_i^t r_{s,i} = \frac{1}{s+1} K^{-1} \sum_{i=1}^K b_i^{t+s+1} \quad (0 \leq s \leq q, s+t \leq m). \quad (1)$$

Put $\delta = \max_{s,i} |r_{s,i} - b_i^{s+1}/(s+1)|$. If

$$\delta \leq c_q(m+1)^{-2q-3}\Delta^{2q+5/2},$$

there is a density f , polynomial of degree at most m on every cell, such that $1/2 \leq f \leq 3/2$ and

$$\int_0^{b_i} a^s f(a) da = r_{s,i} \quad (0 \leq s \leq q), \quad \int f F = \int F \quad (F \in V_m).$$

Consequently a variable of density f and the equal-weight variable on the nodes b_i reproduce all ordered polynomial moments through degree m of a uniform variable and that same node variable. If the node variable integrates a target density w through degree $m+1$, these are also the ordered moments of independent uniform and w variables.

Proof. First prescribe the desired moments through degree q separately on each cell. The difference from the uniform cell moment is at most 2δ in each raw monomial. On a cell $I = [a, a + \ell]$, expand in the shifted Legendre basis of degrees $0, \dots, q$, orthonormal for the uniform probability measure on the cell. Each of its polynomials has raw monomial coefficients bounded by $C_q \ell^{-q}$, since $a \in [0, 1]$ and $\ell \leq 1$. The corresponding moment corrections therefore have size at most $C_q \delta \ell^{-q}$. Their reproducing polynomial density, with the normalizing factor $1/\ell$, is bounded by $C_q \delta \ell^{-q-1}$. We obtain a piecewise polynomial g of degree q with all the prescribed prefix moments and

$$\|g - 1\|_\infty \leq C_q \delta \Delta^{-q-1}.$$

The endpoint conditions and (1) say precisely that $g - 1$ is orthogonal to $\mathcal{S}_q$: for its step generators use Fubini to get $\int g a^s S_t = K^{-1} \sum_i b_i^t r_{s,i}$.

On $W = (I - P_q)V_m$, define $\Lambda((I - P_q)F) = \int (g - 1)F$. Lemma 1 makes this well defined and bounds its norm by $C_q(m+1)^{2q+2}\Delta^{-q-1}\|g - 1\|_2$. Let $h \in W$ be its Riesz representative. It has zero moments through q in every cell and is polynomial of degree at most m there. The cellwise polynomial evaluation estimate yields

$$\|h\|_\infty \leq C(m+1)\Delta^{-1/2}\|h\|_2 \leq C_q(m+1)^{2q+3}\Delta^{-2q-5/2}\delta.$$

Thus $f = g - h$ lies between $1/2$ and $3/2$ for sufficiently small c_q , retains all prescribed cell moments, and integrates every $F \in V_m$ exactly as Lebesgue measure does. For $r + s \leq m$ this says

$$\mathbb{E}[A^r B^s \mathbf{1}_{A < B}] = \int f(a) a^r S_s(a) da = \frac{1}{r+1} K^{-1} \sum_i b_i^{r+s+1}.$$

The ordinary and opposite-chamber moments follow as usual. $\square$

Scope. For each fixed q , all loss factors are polynomial in m and Δ^{-1} . The compatibility data still have to be constructed, and polynomial common denominators have to be maintained across all cells of any proposed induction. This lemma alone does not establish an arbitrary fixed number of polynomial-size finite dice.